Year 10

Mini Test 5 Pathway 1

Finance and Trigonometry

Baldivis Secondary College

Resources allowed: Calculator and 1 x A4 page one side

Name: Solutions. Class: ______ Time: 55 minutes Total Marks: ____/46

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

1. [6 marks: 1, 1, 1, 1, 1, 1]

Circle the correct answer.

a. Find the simple interest on \$1500 at 8% p.a. over 4 years.

A \$360 B \$480

C \$600 D \$720

b. Find the growth factor for 6% interest, compounded annually.

A 0.06 B 1.06

C 0.005 D 1.005

c. If Jennifer borrows \$2000 and has to pay back \$2400, what is the percentage rate of interest?

A 1.7% B 20%

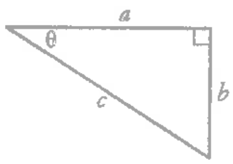
C 40% D 120%

d. What is the value of tan 180°?

A 0 B 1

C $^{-}1$ D Undefined

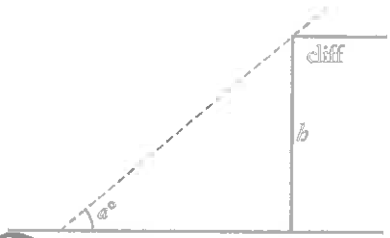
e. In the triangle below, which ratio represents sin θ?



A $\frac{a}{b}$ B $\frac{a}{c}$

C $\frac{b}{c}$ D $\frac{b}{a}$

f. What is the angle *a* a measure of?



A The angle of elevation

B The angle of depression

C The line of sight

D The gradient

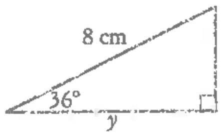
2. [1 mark]

How much will you receive if \$10 000 is invested at 12% interest, compounded annually for 6 years?

A = \$19 738.23 ✓

34. [7 marks: 2, 2, 3]
Find the size of the unknown quantities. Round to 2 d.p.

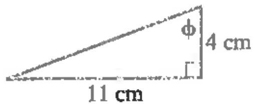
a.



$\cos 36° = \frac{y}{8}$ ✓

$y = 6.47$ ✓

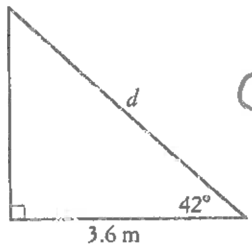
b.



$\tan \phi = \frac{11}{4}$ or 2.75 ✓

$\phi = 70.00°$ ✓

c.



$\cos 42° = \frac{3.6}{d}$ ✓

$d = \frac{3.6}{\cos 42°}$ ✓

$d = 4.84$ ✓

4. [5 marks: 1, 2, 2]
Bert borrowed $1200 until pay day 3 weeks later, when he had to pay back $1290.

a) How much interest did he pay?

$90 ✓

b) If his loan was for a year, how much interest would he have to pay at the same rate?

1 week → $90 \div 3 = \$30$ ✓

Year → $30 \times 52 = \$1560$ ✓

c) What was the per annum interest rate?

Interest rate $= \frac{1560}{1200} \times 100$ ✓

$= 130\%$ ✓

5. [4 marks]

A certain savings account pays 2.5% on minimum monthly balances. Find the interest earned in the following 6 months if the balances are shown in the table below.

Month	Jan	Feb	Mar
Balance	$245.24	$302.56	$128.12
Month	Apr	May	Jun
Balance	$250.30	$276.80	$428.11

2.5% of $245.24 ≃ $6.13
2.5% of $302.56 ≃ $7.56
2.5% of $128.12 ≃ $3.20 (½ each = 3 marks)
2.5% of $250.30 ≃ $6.01
2.5% of $276.80 ≃ $6.92
2.5% of $428.11 ≃ $10.70

Total interest = $40.52 ✓

6. [7 marks: 1, 1, 1, 4]

Mark has been saving his money, so far he has $5500 saved up. He plans to continue saving for another two years and then spend the money on a new car. He decides to invest the $5500 in a term deposit for the two years, so that he can earn some interest on his money. He needs to choose one of the following banks:

BSC Bank - pays 2.5% pa interest compounded annually.

A = $5 778.44 ✓

MD Bank – pays 2.45% pa interest compounded quarterly.

A = $5 781.09 ✓

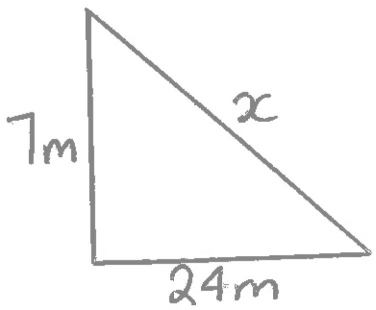
RP Bank – pays 2.4% pa interest compounded monthly.

A = $5 781.69 ✓

By calculating the amount of interest he will earn, decide which bank Mark should invest his money with?

Interest BSC = 5778.44 - 5500 = $278.44 ✓
Interest MD = 5781.09 - 5500 = $281.09 ✓
Interest RP = 5781.69 - 5500 = $281.69 ✓
Mark should invest with RP bank ✓

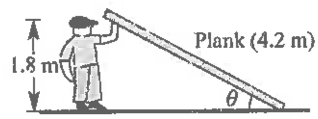
7. [2 marks]
A flagpole broke in a storm. 7 metres are still sticking straight out of the ground, where it snapped, but the remaining piece has hinged over and touches the ground at a point 24 metres away horizontally. How tall was the flagpole before it broke?



$x = \sqrt{7^2 + 24^2}$
$= 25m$ ✓

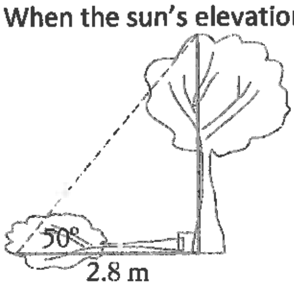
Flagpole $= 25 + 7$
$= 32m$ ✓

8. [2 marks]
Adam, who is 1.8 m tall, holds up a plank of wood 4.2 m long. Find the angle that the plank makes with the ground, correct to 2 decimal places.



$\theta = \sin^{-1} \frac{1.8}{4.2}$ ✓
$= 25.38°$ ✓

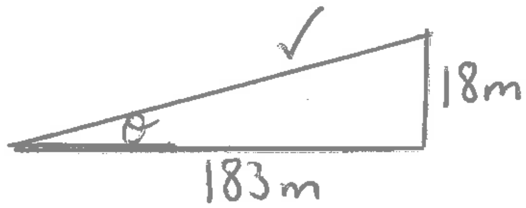
9. [3 marks]
When the sun's elevation in the sky is 50°, a tree casts a shadow 2.8 m long. What is the height of the tree?



$\tan 50° = \frac{h}{2.8}$ ✓
$h = 3.3m$ high ✓

The tree is about 3.3m high ✓

10. [3 marks]
An airplane takes off 183 metres in front of a 18 metre building. At what angle of elevation must the plane take off in order to avoid crashing into the building? Assume that the airplane flies in a straight line and the angle of elevation remains constant until the airplane flies over the building.



$\theta = \tan^{-1} \frac{18}{183}$ ✓
$= 5.6°$ ✓ or $\simeq 6°$

11. [6 marks: 3, 3]

Electronic instruments on a treasure-hunting ship detect a large object on the sea floor. The angle of depression is 29°, and the instruments indicate that the direct-line distance between the ship and the object is about 427 m.

a) How far below the surface of the water is the object?

$x = 427 \times \sin 29^\circ$ ✓

$= 207.01\,m$ ✓✓

b) How far must the ship travel to be directly over it?

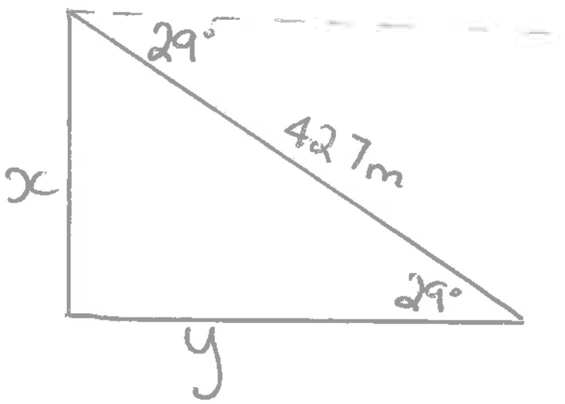
$y = 427 \times \cos 29^\circ$ ✓

$y = 373.46\,m$ ✓✓

OR

$y = \sqrt{427^2 - 207.01^2}$ ✓

$= 373.46\,m$ ✓✓



~End of Test~